

Exercises learning task 15

2026 – 2027

Exercise 9.1: State whether each of the following situations is a correctly stated hypothesis testing problem and why.

- (a) $H_0 : \mu = 25, H_1 : \mu \neq 25$
- (b) $H_0 : \sigma > 10, H_1 : \sigma = 10$
- (c) $H_0 : \bar{x} = 50, H_1 : \bar{x} \neq 50$
- (d) $H_0 : p = 0.1, H_1 : p = 0.5$
- (e) $H_0 : s = 20, H_1 : s < 20$

Exercise 9.5: A textile fiber manufacturer is investigating a new drapery yarn, which the company claims has a mean thread elongation of 12 kg with a standard deviation of 1.5 kg. The company wishes to test the hypothesis $H_0 : \mu = 12$ against $H_1 : \mu < 12$, using a random sample of four specimens.

- (a) What is the type I error probability if the critical region is defined as $\bar{x} < 11.5$ kg?
- (b) Find β for the case in which the true mean elongation is 11.25 kg.
- (c) Find β for the case in which the true mean is 11.5 kg.

Exercise 9.7: In Exercise 9.5, find the boundary of the critical region if the type I error probability is

- (a) $\alpha = 0.01$ and $n = 5$
- (b) $\alpha = 0.05$ and $n = 5$
- (c) $\alpha = 0.01$ and $n = 20$
- (d) $\alpha = 0.05$ and $n = 20$

Exercise 9.9: In Exercise 9.5, calculate the p -value if the observed statistic is

- (a) $\bar{x} = 11.25$
- (b) $\bar{x} = 11.0$
- (c) $\bar{x} = 11.75$

Exercise 9.15: A consumer products company is formulating a new shampoo and is interested in foam height (in millimeters). Foam height is approximately normally distributed and has a standard deviation of 25 mm. The company wishes to test $H_0 : \mu = 175$ mm versus $H_1 : \mu > 175$ mm, using the results of $n = 15$ samples.

- (a) Find the type I error probability α if the critical region is $\bar{x} > 185$.
- (b) What is the probability of type II error if the true mean foam height is 185 mm?
- (c) Find β for the true mean of 195 mm.

Exercise 9.17: In Exercise 9.15, find the boundary of the critical region if the type I error probability is

- (a) $\alpha = 0.01$ and $n = 10$

Exercise 9.19: In Exercise 9.15, calculate the p -value if the observed statistic is

- (a) $\bar{x} = 180$
- (b) $\bar{x} = 190$
- (c) $\bar{x} = 170$

Exercise 9.31: A hypothesis will be used to test that a population mean equals 4 against the alternative that the population mean is more than 4 with known variance σ . What is the

critical value for the test statistic Z_0 for the following significance levels?

(a) 0.01

(b) 0.05

(c) 0.10

Exercise 9.33: For the hypothesis test $H_0 : \mu = 6$ against $H_1 : \mu \neq 6$ and variance known, calculate the p -value for each of the following test statistics.

(a) $z_0 = 2.05$

(b) $z_0 = -1.84$

(c) $z_0 = 0.4$